THE INFINITESIMAL LIFTING CONDITION FOR FORMALLY ETALE MORPHISMS OF ALGEBRAIC SPACES NEEDS NO AFFINENESS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3545-3561 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 3563-3573, the complete author proof (9 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Formally etale morphisms of algebraic spaces and the affine form of the infinitesimal lifting condition; the gluing is retained.

Source reference. Original lemma definition-formally-etale: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3491–3496. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-formally-unramified-not-affine: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2766–2781. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a formally étale morphism of algebraic spaces over S. Then given any solid commutative diagram*

$$\begin{array}{ccc} X & \xleftarrow{a} & T \\ {\scriptstyle f}\downarrow & \nwarrow\ \searrow & \downarrow{\scriptstyle i} \\ Y & \xleftarrow{} & T' \end{array}$$

where $T \subset T'$ is a first order thickening of algebraic spaces over Y there exists exactly one dotted arrow making the diagram commute. In other words, in Definition definition-formally-etale the condition that T be affine may be dropped.

Proof. Let $U' \to T'$ be a surjective étale morphism where $U' = \coprod U'_i$ is a disjoint union of affine schemes. Let $U_i = T \times_{T'} U'_i$. Then we get morphisms $a'_i : U'_i \to X$ such that $a'_i|_{U_i}$ equals the composition $U_i \to T \to X$. By uniqueness (see Lemma lemma-formally-unramified-not-affine) we see that a'_i and a'_j agree on the fibre

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product $U'_i \times_{T'} U'_j$. Hence $\coprod a'_i : U' \to X$ descends to give a unique morphism $a' : T' \to X$. $\square$